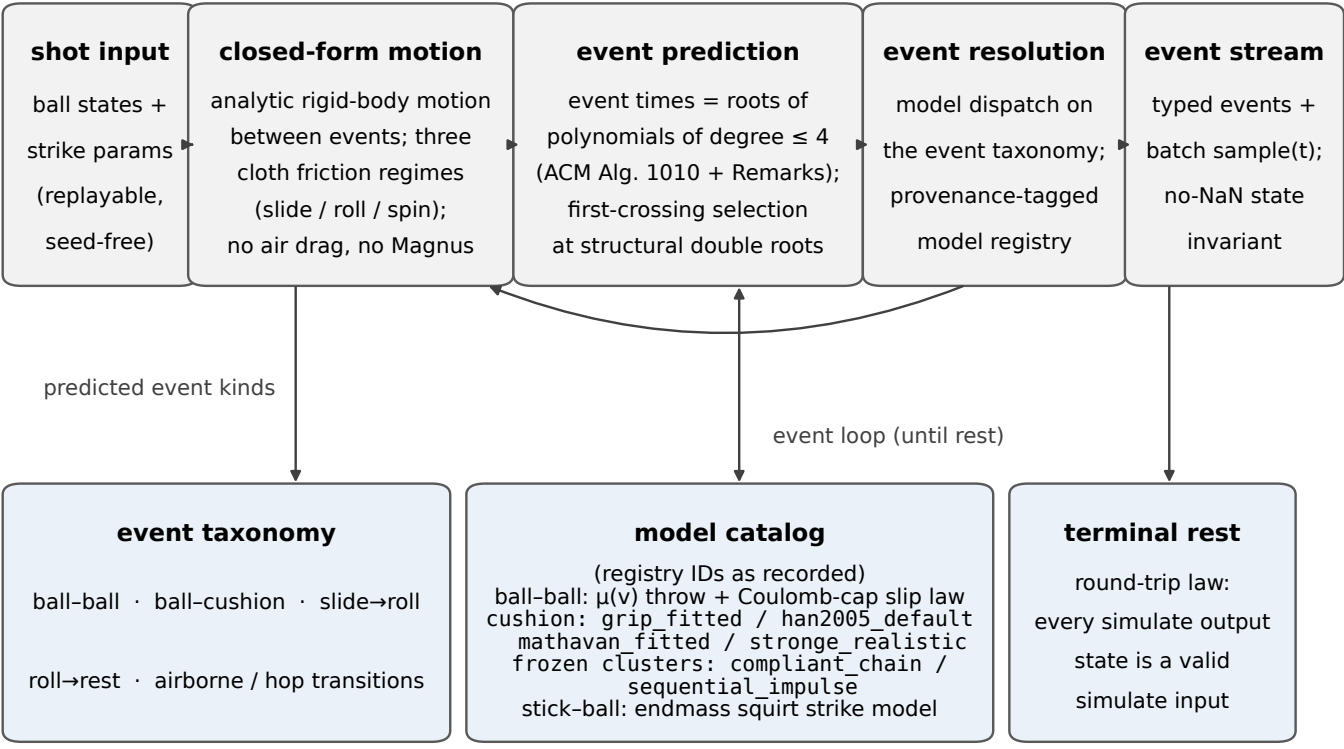
shot input
ball states + strike params (replayable, seed-free)
closed-form motion
analytic rigid-body motion between events; three cloth friction regimes (slide / roll / spin); no air drag, no Magnus
event prediction
event times = roots of polynomials of degree ≤ 4 (ACM Alg. 1010 + Remarks); first-crossing selection at structural double roots
event resolution
model dispatch on the event taxonomy; provenance-tagged model registry
event stream
typed events + batch sample(t); no-NaN state invariant
predicted event kinds
event loop (until rest)
event taxonomy
ball–ball · ball–cushion · slide→roll
roll→rest · airborne / hop transitions
model catalog
(registry IDs as recorded)
ball–ball: μ(v) throw + Coulomb-cap slip law
cushion: grip_fitted / han2005_default
mathavan_fitted / stronge_realistic
frozen clusters: compliant_chain / sequential_impulse
stick–ball: endmass squirt strike model
terminal rest
round-trip law: every simulate output state is a valid simulate input